NG_transmission_emissions_r1.R

June 20, 2024

Contents

`Transmission` *Create gridded natural gas transmission methane emissions maps*

Description

'Transmission' writes 2 netcdf files of gridded methane emissions from natural gas transmission, as well as optional visuals and 2 optional csv files.

Usage

```
Transmission(
  GHGI_file,
  GHGI_Emissions_sheet,
  GHGI_Activity_sheet,
  domain,
  state_name_list,
  output_directory,
  inventory_year,
  verbose,
  plot_directory,
  County_Tigerlines,
  State_Tigerlines,
  focus_city_tigerlines
)
```

Arguments

`GHGI_file` Character providing the full filepath to the GHGI Annex 3.6 excel file. This data is available at https://www.epa.gov/ghgemissions/natural-gas-and-petroleum-systems-gh for the 2022 GHGI. In the GHGI Annexes, available at https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks-1990-2020, there is a link to the file in Section 3.6: "Methodology for Estimating CH4, CO2, and N2O Emissions from Natural Gas Systems". The excel file has multiple sheets, each of which has a separate layout. There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

`GHGI_Emissions_sheet`
: Character providing the sheet name in "GHGI_file" that provides the "CH4 Emissions for Natural Gas Systems, by Segment and Source, for All Years". The sheet name as of the 2022 GHGI is "3.6-1".

`GHGI_Activity_sheet`
: Character providing the sheet name in "GHGI_file" that provides the "Activity Data for Natural Gas Systems Sources, for All Years". The sheet name as of the 2022 GHGI is "3.6-7".

`domain`
: SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

`state_name_list`
: Character vector listing all states within the desired domain

`output_directory`
: Character providing the full filepath to save processed data

`inventory_year`
: Character indicating the desired year of data to use.

`verbose`
: Logical indicating whether to save additional output. This includes plots of the gridded methane emissions for each fuel-sector-inventory-variation combination as well as 2 summed plots for each inventory-variation combination - one for wood and one for all other sectors.

`plot_directory`
: Character providing the full filepath to save figures. Only relevant if verbose = TRUE.

`County_Tigerlines`
: SpatVector. United States Census Bureau county shapefile downloaded in Main.

`State_Tigerlines`
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if verbose=TRUE.

`focus_city_tigerlines`
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if a focus city was set in main and verbose=TRUE.

Details

This function calculates and grids methane emissions from natural gas transmission systems. It uses the Homeland Infrastructure Foundation-Level Data (HIFLD), Environmental Protection Agency's (EPA) Greenhouse Gas Inventory (GHGI), EPA Greenhouse Gas Reporting Program (GHGRP) and Energy Information Administration (EIA) Energy Atlas data. First the relevant data is pulled from the GHGI Annex file.

For pipelines the GHGI data includes the emissions and activity data for pipeline leaks, meter and regulating stations, and venting. An emission factor in mols of methane per meter of pipeline per second can then be calculated. This emission factor is then applied to the EIA inter/intrastate pipeline data.

For compressors the GHGI data includes the emissions and activity data for compressor station fugitive emissions, dehydrator vents, flaring, engines, turbines, generators, pneumatic devices, and venting. For generators they are split into engine and turbine emissions so the ratio between transmission/storage emissions for engines and turbines are applied to the generator values to get the transmission component of generator emissions. An national average emission rate in mols of methane per station per second can then be calculated. This emission rate is then assigned to all HIFLD compressors. GHGRP data is then used to overwrite this default emission rate. Note most compressor stations do not report their emissions to the GHGRP, so only a subset will be overwritten using GHGRP data. The GHGRP compressor emissions are scaled so that the average emissions within the domain are equal to the national average calculated from the GHGI. As their is not a common identifier between the GHGRP and HIFLD datasets, the nearest facility is considered the matching facility, though an error is flagged if a GHGRP compressor has no matching HIFLD compressor within 1 km.

The necessary GHGRP compressor location/emissions data, EIA pipeline data and HIFLD compressor location data will be automatically downloaded.

The GHGRP includes only facilities that emit at least 25,000 metric tons of carbon dioxide equivalent while the GHGI is intended to capture all national emissions. GHGRP data is available starting in 2010 and generally is about 2 years behind present day, the GHGI is available starting in 1990 and is updated approximately in sync with the GHGRP. Both datasets are annual. The GHGRP is at the facility scale while the GHGI is national totals.

The EIA pipeline data being used was last updated January 2020. The HIFLD compressor data being used was last updated December 2022.

The GHGI is available at `https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-s` the GHGRP is available at `https://ghgdata.epa.gov/ghgp/main.do`, the HIFLD data is available at `https://hifld-geoplatform.hub.arcgis.com/datasets/geoplatform::natural-gas-compressor-stat` `about`, and the EIA data is available at `https://atlas.eia.gov/datasets/eia::natural-gas-interstate-and-in` `about`.

Value

Nothing is returned from the function, but the main outputs are 2 netcdf files of the methane emissions from natural gas transmission. They are titled "NG_trans_compressors.nc" and "NG_trans_pipes.nc" where NG=natural gas and trans=transmission. The first file is for transmission compressor emissions and the second is for transmission pipeline emissions.

2 csv files are also optionally saved. These are "NG_trans_compressors.csv" and "NG_trans_compressors_all.csv". The simpler csv includes only the name, location, and assigned emissions for compressors within the domain that were pulled from the GHGRP The _all files include all variables that were in the corresponding input file for the same compressors.

Author(s)

Joe Pitt, <madeup@wisc.edu>

Kris Hajny, <blank@fake.edu>

Israel Lopez-Coto, <test@test.edu>

Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
            ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
```

```
            xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
            crs=grid_crs)
Urban_Tigerlines <- vect("~/../Desktop/Urban_Tigerlines/tl_2018_us_uac10.shp")
focus_city <- terra::subset(Urban_Tigerlines,Urban_Tigerlines$NAME10 %in% "Philadelphia, PA--NJ--DE--MD")
Transmission(GHGI_file="~/../Desktop/2022_ghgi_natural_gas_systems_annex36_tables.xlsx",
                     GHGI_Emissions_sheet="3.6-1",
                     GHGI_Activity_sheet="3.6-7",
                     domain=grid,
                     state_name_list=c("DE","MD","NJ","NY","PA"),
                     output_directory="~/../Desktop/",
                     inventory_year=2018,
                     verbose=TRUE,
              State_Tigerlines=vect("~/../Desktop/State_Tigerlines/tl_2018_us_state.shp"),
              County_Tigerlines=vect("~/../Desktop/County_Tigerlines/tl_2018_us_county.shp"),
                     focus_city_tigerlines=focus_city,
                     plot_directory="~/../Desktop/plots/")
```

Index